

Integral, Transform, and Fractional-Calculus Frontiers for the Fabius–Rvachev System

Exact kernel formulae, inverse-quantile calculus, Cauchy renormalization,
and a Jacobi continued fraction for Rvachev’s up-law

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Recursive corpus audit anchored at commit 32d6d36c51d803289e6d6a0dc0c37753766eba47,
with current living manuscripts and active frontier drafts rechecked on the same date

Abstract

The Fabius–Rvachev documentation corpus already contains a mature theory of the Thue–Morse sequence, the bounded Fabius function, Rvachev’s compactly supported up-function, exact dyadic values, repeated integration, infinite sinc products, q -binomial acceleration, and lower-Lambert endpoint asymptotics. This report performs a non-duplication audit of that recursive LaTeX corpus and develops an integral calculus that joins several structures not previously assembled there.

The main exact device is probabilistic. If $X = \sum_{k \geq 1} 2^{-k} U_k$, where the U_k are independent uniform variables on $[0, 1]$, then the bounded Fabius function F is the distribution function of X , while on $[0, 1]$, $\text{up}(x) = 1 - F(x) = \mathbb{P}(X > x)$. This converts weighted integrals of F and up into expectations of primitives and truncated powers. It yields closed forms for monomial weights, exponential kernels, Mellin transforms, all integer repeated antiderivatives, and fractional Riemann–Liouville integrals. For $p > -1$ and $\alpha > 0$, the fractional antiderivatives of $x^p F(x)$ and $x^p \text{up}(x)$ are expressed by incomplete beta kernels; at the endpoint they become convergent series in generalized moments $m(s) = \mathbb{E}X^s$. Reflection of the Fabius law gives an entire Newton series $m(s) = \sum_{n \geq 0} (-1)^n \binom{s}{n} m_n$, thereby covering fractional and negative powers by rational integer moments.

For the inverse Fabius function $Q = F^{-1}$, a layer-cake formula gives its ordinary and fractional antiderivatives, Laplace/Fourier transforms, mixed Mellin identities, and all quantile moments. A particularly direct bridge is

$$\int_0^1 \text{up}(x)^\alpha dx = \Gamma(\alpha + 1) (I_{0+}^\alpha Q)(1).$$

For the full up-law, the Cauchy transform $S(z) = \int_{-1}^1 \text{up}(t)/(z - t) dt$ satisfies the new exact functional-differential equation $S'(z) = 2[S(2z + 1) - S(2z - 1)]$. Combining this with the rational moment sequence resolves a continued-fraction direction explicitly posed in an active repository draft: S has a symmetric Jacobi fraction with positive rational coefficients, whose first values are computed exactly and whose limit is $1/4$. Finally, nonlinear power integrals are identified with order statistics and shown to obey $\int_0^1 \text{up}(x)^p dx \sim Q(1/p)$.

All theorems designated as proved are supplied with arguments. Numerical work is used only as an independent check and is reproduced by the commented Python program included with the report. Novelty claims are relative to the audited repository, not assertions of exhaustive worldwide priority.

Status discipline. **Background / corpus result.** marks facts already present in the repository or standard literature. **Proved in this report.** marks deductions proved here and not located in the audited corpus in the displayed form. **Conditional statement.** requires an explicit regularity hypothesis. **Conjectural.** is an open proposal. Numerical agreement never substitutes for proof.

Contents

1	Corpus audit and the non-duplication boundary	4
1.1	Coverage protocol	4
1.2	What is imported and what is added	4
2	Normalizations and the probabilistic platform	5
2.1	The bounded Fabius law and the up-law	5
2.2	Entire exponential products and rational cumulants	5
3	The universal survival-kernel calculus	6
3.1	Arbitrary kernels and partial antiderivatives	6
3.2	Exponential kernels and ordinary transforms	7
3.3	Derivative-weighted beta integrals	7
4	Generalized moments and the Mellin transform	8
4.1	An entire Newton series from reflection	8
4.2	A three-function Mellin dictionary	9
5	Inverse Fabius integral calculus	9
5.1	Area identity and ordinary antiderivatives	9
5.2	A general layer-cake transform	9
5.3	Laplace, Fourier, and mixed Mellin transforms of the inverse	10
5.4	All ordinary derivatives of the inverse	10
6	Integer-order derivative and antiderivative hierarchies	11
6.1	The Thue–Morse derivative tower	11
6.2	All repeated antiderivatives as truncated moments	11
7	Fractional calculus	12
7.1	Conventions	12
7.2	Fractional primitives and derivatives of the Fabius function	12
7.3	Weighted fractional antiderivatives: the incomplete-beta master theorem	13
7.4	Fractional primitives of the up-function	13
7.5	Fractional calculus of the inverse Fabius function	14
7.6	A full-line Riesz refinement law	14
8	Cauchy, Stieltjes, and Hilbert transforms	14
8.1	The Cauchy transform and its representations	14
8.2	An exact functional-differential equation	15
8.3	Boundary values and the Hilbert transform	15
9	The Jacobi continued fraction of the up-law	16
9.1	From rational moments to a J-fraction	16
10	Nonlinear powers and order statistics	17
10.1	Exact order-statistic identities	17
10.2	Large-power asymptotics from inverse slow variation	18
11	Correlation and transform families	19

12 Numerical verification and reproducibility	20
12.1 Exact and floating-point layers	20
12.2 Reproduction command	21
13 Conjectures and future research directions	21
13.1 Cauchy renormalization as a spectral equation	21
13.2 Jacobi arithmetic and endpoint flatness	21
13.3 Fractional endpoint asymptotics	22
13.4 Nonlinear power integrals	22
13.5 Fractional formalization roadmap	22
14 Conclusion	22
A Exact formula sheet	23
B Corpus families inspected	23
C Reproducibility manifest	24

1 Corpus audit and the non-duplication boundary

1.1 Coverage protocol

The directory `Analysis/FabiusFunction/docs` is a layered research environment rather than a single paper. At the pinned 27 August 2026 snapshot, an existing audit inventory records all 67 distinct `*.tex` paths. The archive README explains that historical files are frozen provenance objects and that byte-identical current duplicates are pruned. The non-formalized frontier directory is explicitly a quarantine boundary: its claims are not silently promoted into the formalized primary exposition [4, 3, 7].

The present audit used four passes.

1. The 67-path inventory supplied the recursive coverage scaffold.
2. Living primary manuscripts, the consolidated frontier volume, the Thue–Morse atlas, inverse-function reports, repeated-integration reports, and every active top-level frontier report were reopened on the current branch.
3. Frozen archive entries were compared by mathematical lineage, so that a superseded copy did not count as an independent novelty witness, while a distinctive historical argument remained part of the boundary.
4. Candidate phrases and neighboring concepts were searched thematically: Riemann–Liouville and Caputo calculus, fractional primitives, inverse layer-cake formulae, Cauchy and Hilbert transforms, Stieltjes/Jacobi fractions, order statistics, power integrals, and generalized negative moments.

This procedure located substantial adjacent work. In particular, the corpus already contains rational up-moments, Hankel determinants, monic orthogonal polynomials, Gaussian and Lobatto quadrature, and repeated-integration hierarchies. One active report explicitly proposes a Jacobi continued fraction for the moment generating function or Cauchy transform as a future direction [5]. Therefore the continued fraction below is presented as a resolution of a recorded gap, not as a topic absent from the research program.

1.2 What is imported and what is added

Theme	Existing corpus foundation	Addition in this report
Probability dictionary	F as a smooth distribution function; up as the density of a symmetric random series; exact Laplace/Fourier products	Universal survival-kernel calculus for arbitrary weights, including partial antiderivatives and truncated moments
Repeated integration	Integer repeated-integral formulae and exact derivative towers	One incomplete-beta theorem interpolating all integer and fractional orders for $x^p F$ and x^p up
Moments and Bernoulli data	Rational integer moments from Bernoulli cumulants and Bell recurrences	Entire Newton interpolation $m(s)$, Mellin continuation, negative powers, and logarithmic integrals
Inverse Fabius	Dyadic inverse germs, Richardson acceleration, endpoint Lambert–Gamma–zeta asymptotics, inverse elasticity	Layer-cake transforms, quantile antiderivatives, mixed Mellin identities, and fractional Q -calculus
Orthogonal polynomials	Hankel norms and positive Gaussian quadrature; J-fraction posed as an open problem	Cauchy renormalization equation, explicit symmetric J-fraction, exact rational coefficients, and the $1/4$ limit
Nonlinear integrals	Endpoint/inverse asymptotics sufficient to prove slow variation of Q	Order-statistic formulae and $\int \text{up}^p \sim Q(1/p)$, with a second-order frontier conjecture
Fractional derivatives	No Riemann–Liouville/Caputo treatment located	Exact fractional refinement equations for F , up, and Q , plus a full-line Riesz refinement law

Remark 1.1 (Priority boundary). The formulae below often use classical probability, Mellin, fractional-calculus, and orthogonal-polynomial tools. The novelty assertion is the corpus-relative combination and the displayed Fabius–Rvachev specializations. A worldwide priority claim would require a dedicated multilingual search using all historical normalizations of the Fabius and up-functions.

2 Normalizations and the probabilistic platform

2.1 The bounded Fabius law and the up-law

Let $U_1, U_2, \dots$ be independent and uniformly distributed on $[0, 1]$, and put

$$X = \sum_{k=1}^{\infty} \frac{U_k}{2^k}. \quad (2.1)$$

Then $0 < X < 1$ almost surely. Let

$$F(x) = \mathbb{P}(X \leq x), \quad 0 \leq x \leq 1, \quad (2.2)$$

and extend it by $F(x) = 0$ for $x < 0$, $F(x) = 1$ for $x > 1$. This is the bounded Fabius function in the normalization used throughout the repository. It is a strictly increasing C^∞ bijection of $[0, 1]$, flat at both endpoints, and satisfies

$$F(1-x) = 1 - F(x), \quad F'(x) = 2 \operatorname{up}(2x-1). \quad (2.3)$$

Let $V_k = 2U_k - 1$, uniform on $[-1, 1]$, and define

$$Y = 2X - 1 = \sum_{k=1}^{\infty} \frac{V_k}{2^k}. \quad (2.4)$$

The density of Y is Rvachev's even up-function up , supported on $[-1, 1]$, normalized by $\int \operatorname{up} = 1$ and $\operatorname{up}(0) = 1$. On the positive half of its support,

$$\boxed{\operatorname{up}(x) = F(1-x) = 1 - F(x) = \mathbb{P}(X > x), \quad 0 \leq x \leq 1.} \quad (2.5)$$

Thus the right half of the up-bump is exactly the survival function of the Fabius probability law.

The fixed-point decompositions

$$X \stackrel{d}{=} \frac{U + X'}{2}, \quad Y \stackrel{d}{=} \frac{V + Y'}{2} \quad (2.6)$$

use independent copies X', Y' . They imply

$$\operatorname{up}'(x) = 2(\operatorname{up}(2x+1) - \operatorname{up}(2x-1)) \quad (2.7)$$

and, on $0 < x < 1/2$, $F'(x) = 2F(2x)$. These are established corpus identities [1, 10].

2.2 Entire exponential products and rational cumulants

Define

$$A(z) = \mathbb{E}e^{zX} = \prod_{k=1}^{\infty} \frac{e^{z/2^k} - 1}{z/2^k}, \quad (2.8)$$

$$\Phi(z) = \mathbb{E}e^{zY} = e^{-z}A(2z) = \prod_{k=1}^{\infty} \frac{\sinh(z/2^k)}{z/2^k}. \quad (2.9)$$

Both are entire. With Bernoulli numbers in the convention $B_1 = -1/2$, the cumulants of X are

$$\kappa_1(X) = \frac{1}{2}, \quad \boxed{\kappa_n(X) = \frac{B_n}{n(2^n - 1)}, \quad n \geq 2,} \quad (2.10)$$

and those of Y are

$$\kappa_1(Y) = 0, \quad \boxed{\kappa_n(Y) = \frac{2^n B_n}{n(2^n - 1)}, \quad n \geq 2.} \quad (2.11)$$

Consequently all integer moments are rational and are generated by complete Bell polynomials:

$$m_n := \mathbb{E}X^n = \mathcal{B}_n(\kappa_1(X), \dots, \kappa_n(X)), \quad \mu_n := \mathbb{E}Y^n = \mathcal{B}_n(\kappa_1(Y), \dots, \kappa_n(Y)). \quad (2.12)$$

The first values are

$$(m_0, m_1, m_2, m_3, m_4, m_5, m_6) = \left(1, \frac{1}{2}, \frac{5}{18}, \frac{1}{6}, \frac{143}{1350}, \frac{19}{270}, \frac{1153}{23814}\right), \quad (2.13)$$

$$(\mu_0, \mu_2, \mu_4, \mu_6, \mu_8) = \left(1, \frac{1}{9}, \frac{19}{675}, \frac{583}{59535}, \frac{132809}{32531625}\right). \quad (2.14)$$

3 The universal survival-kernel calculus

3.1 Arbitrary kernels and partial antiderivatives

Theorem 3.1 (Survival-kernel identity). *Let $k \in L^1(0, 1)$, and put $K(x) = \int_0^x k(t) dt$. Then*

$$\boxed{\int_0^1 k(t) \text{up}(t) dt = \mathbb{E}K(X).} \quad (3.1)$$

More generally, for $0 \leq x \leq 1$,

$$\boxed{\int_0^x k(t) \text{up}(t) dt = \mathbb{E}K(\min\{X, x\}).} \quad (3.2)$$

Both identities remain valid for complex-valued kernels whenever the displayed integrals are absolutely convergent. **Proved in this report.**

Proof. By (2.5), $\text{up}(t) = \mathbb{E}\mathbf{1}_{\{t < X\}}$. Tonelli–Fubini gives

$$\int_0^x k(t) \text{up}(t) dt = \mathbb{E} \int_0^x k(t) \mathbf{1}_{\{t < X\}} dt = \mathbb{E} \int_0^{\min(X, x)} k(t) dt.$$

Taking $x = 1$ proves (3.1). □

Corollary 3.2 (Monomial antiderivatives). *For $\Re p > -1$ and $0 \leq x \leq 1$,*

$$\int_0^x t^p \text{up}(t) dt = \frac{\mathbb{E} \min(X, x)^{p+1}}{p+1}, \quad (3.3)$$

$$\int_0^x t^p F(t) dt = \frac{x^{p+1} - \mathbb{E} \min(X, x)^{p+1}}{p+1}. \quad (3.4)$$

At $x = 1$, writing $m(s) = \mathbb{E}X^s$,

$$\boxed{\int_0^1 t^p \text{up}(t) dt = \frac{m(p+1)}{p+1},} \quad (3.5)$$

$$\boxed{\int_0^1 t^p F(t) dt = \frac{1 - m(p+1)}{p+1}.} \quad (3.6)$$

Proved in this report.

For nonnegative integers, this gives for example

	$p = 0$	$p = 1$	$p = 2$
$\int_0^1 x^p \text{up}(x) \, dx$	1/2	5/36	1/18
$\int_0^1 x^p F(x) \, dx$	1/2	13/36	5/18

Remark 3.3 (Negative powers). The survival integral in (3.5) has the sharp local restriction $\Re p > -1$, because $\text{up}(0) = 1$. In contrast, F is flat at zero, so $\int_0^1 x^p F(x) \, dx$ converges for every $p \in \mathbb{C}$; the right side of (3.6) supplies its entire continuation after removing the apparent singularity at $p = -1$. In particular,

$$\int_0^1 \frac{F(x)}{x} \, dx = -m'(0) = \mathbb{E}[-\log X]. \quad (3.7)$$

3.2 Exponential kernels and ordinary transforms

Taking $k(t) = e^{zt}$ in theorem 3.1 yields the entire identities

$$\boxed{\int_0^1 \text{up}(t) e^{zt} \, dt = \frac{A(z) - 1}{z}}, \quad (3.8)$$

$$\boxed{\int_0^1 F(t) e^{zt} \, dt = \frac{e^z - A(z)}{z}}, \quad (3.9)$$

with removable values at $z = 0$. Differentiation under the integral gives all monomial-exponential combinations:

$$\int_0^1 t^n \text{up}(t) e^{zt} \, dt = \frac{d^n}{dz^n} \left(\frac{A(z) - 1}{z} \right), \quad (3.10)$$

$$\int_0^1 t^n F(t) e^{zt} \, dt = \frac{d^n}{dz^n} \left(\frac{e^z - A(z)}{z} \right). \quad (3.11)$$

At $z = -2\pi i\xi$, these are the finite-interval Fourier transforms of F and the positive half of up .

3.3 Derivative-weighted beta integrals

Theorem 3.4 (Probability-integral-transform family). *Let $a, b \in \mathbb{C}$ with $\Re a, \Re b > -1$. Then*

$$\boxed{\int_0^1 F(x)^a (1 - F(x))^b F'(x) \, dx = B(a + 1, b + 1)}. \quad (3.12)$$

More generally, for any integrable φ ,

$$\int_0^1 \varphi(F(x)) F'(x) \, dx = \int_0^1 \varphi(u) \, du. \quad (3.13)$$

Equivalently, since $F'(x) = 2 \text{up}(2x - 1)$, this is an exact family of integrals mixing the up -function with powers of the Fabius function. **Proved in this report.**

Proof. Use the substitution $u = F(x)$, legitimate because F is a smooth strictly increasing bijection of $[0, 1]$. $\square$

The quantile-weighted extension is

$$\int_0^1 x^r F(x)^a (1 - F(x))^b F'(x) \, dx = \int_0^1 Q(u)^r u^a (1 - u)^b \, du, \quad (3.14)$$

where $Q = F^{-1}$. Thus every beta-weighted moment of the inverse is a forward Fabius integral.

4 Generalized moments and the Mellin transform

4.1 An entire Newton series from reflection

The flat endpoints imply more than the existence of all negative moments.

Lemma 4.1 (Super-polynomial decay of integer moments). *For every $N > 0$,*

$$m_n = \mathbb{E}X^n = \mathcal{O}_N(n^{-N}) \quad (n \rightarrow \infty). \quad (4.1)$$

Proved in this report.

Proof. By reflection, $\mathbb{P}(X > 1 - t) = F(t)$. Endpoint flatness gives $F(t) = \mathcal{O}_M(t^M)$ for every M . The layer-cake formula

$$m_n = n \int_0^1 x^{n-1} \mathbb{P}(X > x) dx$$

followed by $x = 1 - t$, a split at fixed small t , and $(1 - t)^{n-1} \leq e^{-(n-1)t}$ gives $m_n = \mathcal{O}_M(n^{-M})$. $\square$

Theorem 4.2 (Entire Newton interpolation of Fabius moments). *For every $s \in \mathbb{C}$,*

$$m(s) := \mathbb{E}X^s = \sum_{n=0}^{\infty} (-1)^n \binom{s}{n} m_n. \quad (4.2)$$

*The series converges absolutely and locally uniformly in s , hence defines an entire function. Its coefficients m_n are rational Bell polynomials in the Bernoulli–Mersenne cumulants (2.10). *Proved in this report.**

Proof. The reflection law gives $1 - X \stackrel{d}{=} X$. For every fixed $0 < X < 1$,

$$X^s = (1 - (1 - X))^s = \sum_{n \geq 0} (-1)^n \binom{s}{n} (1 - X)^n.$$

On compact subsets of the s -plane, the generalized binomial coefficients grow at most polynomially in n , while [theorem 4.1](#) makes m_n smaller than every inverse power. Thus expectation and summation may be interchanged, and local uniform convergence follows. $\square$

Important specializations are

$$m(-r) = \sum_{n=0}^{\infty} \binom{n+r-1}{n} m_n, \quad r \in \mathbb{N}, \quad (4.3)$$

$$-m'(0) = \sum_{n=1}^{\infty} \frac{m_n}{n}, \quad (4.4)$$

so (3.7) has a convergent series of rational terms. Numerically,

$$\int_0^1 \frac{F(x)}{x} dx = 0.7582399961 \dots, \quad \int_0^1 \frac{F(x)}{x^2} dx = m(-1) - 1 = 1.314885 \dots \quad (4.5)$$

The decimals are checks, not claimed closed constants.

4.2 A three-function Mellin dictionary

Theorem 4.3 (Mellin transform triad). *For $\Re s > 0$,*

$$\mathcal{M}[\text{up}|_{[0,1]}](s) = \int_0^1 x^{s-1} \text{up}(x) \, dx = \frac{m(s)}{s}, \quad (4.6)$$

$$\mathcal{M}[F](s) = \int_0^1 x^{s-1} F(x) \, dx = \frac{1 - m(s)}{s}, \quad (4.7)$$

$$\mathcal{M}[F'](s) = \int_0^1 x^{s-1} F'(x) \, dx = m(s-1). \quad (4.8)$$

*The first extends meromorphically to $\mathbb{C}$ with one simple pole at $s = 0$, of residue one. The second and third extend entire. **Proved in this report.***

Proof. The first identity is (3.5) with $p = s - 1$. The second follows from $F = 1 - \text{up}$, and the third is the definition of expectation under the density F' . Entire continuation uses theorem 4.2; the zero of $1 - m(s)$ at $s = 0$ removes the apparent pole in (4.7). $\square$

The Mellin triad isolates a striking analytic asymmetry. The positive half of the up-function equals one at the origin and therefore contributes the single Mellin pole. The Fabius function and its density are flat there, and their Mellin transforms are entire.

5 Inverse Fabius integral calculus

Let

$$Q = F^{-1} : [0, 1] \rightarrow [0, 1]. \quad (5.1)$$

It satisfies $Q(1 - u) = 1 - Q(u)$.

5.1 Area identity and ordinary antiderivatives

Theorem 5.1 (Inverse-area antiderivative). *For $0 \leq u \leq 1$,*

$$\boxed{\int_0^u Q(v) \, dv = uQ(u) - \int_0^{Q(u)} F(x) \, dx.} \quad (5.2)$$

*In particular, $\int_0^1 Q(u) \, du = 1/2$. **Proved in this report.***

Proof. The rectangle $[0, Q(u)] \times [0, u]$ is partitioned, up to the graph, into the area below F and the area to the left of Q . Equivalently, substitute $v = F(x)$ and integrate by parts. $\square$

5.2 A general layer-cake transform

Theorem 5.2 (Forward-inverse layer cake). *Let $\phi \in L^1(0, 1)$, and let Ψ be absolutely continuous with $\Psi(0) = 0$. Whenever the right side is absolutely integrable,*

$$\boxed{\int_0^1 \phi(u) \Psi(Q(u)) \, du = \int_0^1 \Psi'(x) \left(\int_{F(x)}^1 \phi(u) \, du \right) \, dx.} \quad (5.3)$$

Proved in this report.

Proof. Write $\Psi(Q(u)) = \int_0^1 \Psi'(x) \mathbf{1}_{\{x < Q(u)\}} \, dx$. Since $x < Q(u)$ is equivalent to $F(x) < u$, Fubini gives the result. $\square$

With $\Psi(x) = x^p$, $\Re p > 0$, this becomes

$$\int_0^1 \phi(u) Q(u)^p \, du = p \int_0^1 x^{p-1} \left(\int_{F(x)}^1 \phi(u) \, du \right) \, dx. \quad (5.4)$$

Taking $\phi = 1$ proves the exact quantile-moment identity

$$\boxed{\int_0^1 Q(u)^p \, du = m(p)}, \quad \Re p > 0, \quad (5.5)$$

with analytic continuation supplied by [theorem 4.2](#) where desired.

5.3 Laplace, Fourier, and mixed Mellin transforms of the inverse

For $s \neq 0$, [theorem 5.2](#) with $\Psi(x) = x$ gives

$$\boxed{\int_0^1 e^{-su} Q(u) \, du = \frac{1}{s} \int_0^1 (e^{-sF(x)} - e^{-s}) \, dx}. \quad (5.6)$$

Similarly, for $t \neq 0$,

$$\boxed{\int_0^1 e^{itu} Q(u) \, du = \frac{1}{it} \int_0^1 (e^{it} - e^{itF(x)}) \, dx}. \quad (5.7)$$

These identities replace a transform of a difficult inverse by nonlinear but forward evaluations of F .

A two-parameter Mellin identity follows by taking $\phi(u) = u^{s-1}$.

Theorem 5.3 (Mixed forward-inverse Mellin identity). *For $\Re s > 0$, $\Re p > 0$,*

$$\boxed{\int_0^1 u^{s-1} Q(u)^p \, du = \frac{1}{s} - \frac{p}{s} \int_0^1 x^{p-1} F(x)^s \, dx}. \quad (5.8)$$

Proved in this report.

This makes nonlinear powers F^s and inverse moments two faces of one transform. For integer s , the right side has the order-statistic interpretation developed in [section 10](#).

5.4 All ordinary derivatives of the inverse

Put $f = F'$, and, after differentiating with respect to the spatial variable x , define

$$R_1(x) = \frac{1}{f(x)}, \quad R_{n+1}(x) = \frac{R'_n(x)}{f(x)}. \quad (5.9)$$

Then

$$\boxed{Q^{(n)}(u) = R_n(Q(u)), \quad 0 < u < 1.} \quad (5.10)$$

This is simply the operator formula

$$Q^{(n)}(u) = \left(\frac{1}{F'(x)} \frac{d}{dx} \right)^{n-1} \frac{1}{F'(x)} \Big|_{x=Q(u)}. \quad (5.11)$$

The first cases are

$$Q' = f^{-1}, \quad Q'' = -\frac{f'}{f^3}, \quad (5.12)$$

$$Q''' = \frac{3(f')^2 - f f''}{f^5}, \quad Q'''' = \frac{-15(f')^3 + 10 f f' f'' - f^2 f'''}{f^7}, \quad (5.13)$$

all evaluated at $x = Q(u)$. The recurrence is preferable to a single large Faà di Bruno polynomial when high orders are required.

6 Integer-order derivative and antiderivative hierarchies

6.1 The Thue–Morse derivative tower

Let $s_2(j)$ denote the binary digit sum. Iterating (2.7) gives the established exact formula

$$\text{up}^{(n)}(x) = 2^{n(n+1)/2} \sum_{j=0}^{2^n-1} (-1)^{s_2(j)} \text{up}(2^n x + 2^n - 1 - 2j). \quad (6.1)$$

On the left edge of the Fabius interval,

$$F^{(n)}(x) = 2^{n(n+1)/2} F(2^n x), \quad 0 \leq x \leq 2^{-n}. \quad (6.2)$$

These are imported corpus results; their role here is to connect the integral formulae to arbitrary derivative order.

For any nonnegative integer m , integration by parts and endpoint flatness give

$$\int_{-1}^1 x^m \text{up}^{(n)}(x) dx = \begin{cases} (-1)^n \frac{m!}{(m-n)!} \mu_{m-n}, & m \geq n, \\ 0, & m < n, \end{cases} \quad (6.3)$$

and, for $n \geq 1$,

$$\int_0^1 x^p F^{(n)}(x) dx = (-1)^{n-1} p^{n-1} m(p-n+1), \quad (6.4)$$

initially where the integration-by-parts calculation is classical and then by analytic continuation. Exponential weights similarly yield

$$\int_{-1}^1 e^{zx} \text{up}^{(n)}(x) dx = (-z)^n \Phi(z), \quad (6.5)$$

$$\int_0^1 e^{zx} F^{(n)}(x) dx = (-z)^{n-1} A(z), \quad n \geq 1. \quad (6.6)$$

6.2 All repeated antiderivatives as truncated moments

For integer $n \geq 1$,

$$(I_{0+}^n F)(x) = \frac{1}{n!} \mathbb{E}(x - X)_+^n, \quad (6.7)$$

$$(I_{-1+}^n \text{up})(x) = \frac{1}{(n-1)!} \mathbb{E}(x - Y)_+^{n-1}, \quad (6.8)$$

$$(I_{0+}^n Q)(u) = \frac{1}{n!} \int_0^{Q(u)} (u - F(x))^n dx. \quad (6.9)$$

The first and third are special cases of fractional theorems below. For $x > 1$, the up antiderivative becomes the finite moment polynomial

$$(I_{-1+}^n \text{up})(x) = \frac{1}{(n-1)!} \sum_{r=0}^{\lfloor (n-1)/2 \rfloor} \binom{n-1}{2r} \mu_{2r} x^{n-1-2r}. \quad (6.10)$$

Thus the rational up-moments are precisely the coefficients of all exterior ordinary antiderivatives.

7 Fractional calculus

7.1 Conventions

For $\alpha > 0$, the left Riemann–Liouville integral is

$$(I_{a+}^{\alpha}g)(x) = \frac{1}{\Gamma(\alpha)} \int_a^x (x-t)^{\alpha-1} g(t) dt. \quad (7.1)$$

For $n-1 < \alpha < n$,

$$D_{a+}^{\alpha}g = \frac{d^n}{dx^n} I_{a+}^{n-\alpha}g, \quad {}^C D_{a+}^{\alpha}g = I_{a+}^{n-\alpha}g^{(n)}. \quad (7.2)$$

These conventions follow the standard monograph of Samko, Kilbas, and Marichev [12]. Because F and up are flat at their relevant left endpoints, the usual boundary correction terms between Riemann–Liouville and Caputo derivatives vanish whenever both sides are classically defined.

7.2 Fractional primitives and derivatives of the Fabius function

Theorem 7.1 (Fractional truncated-power formula for F). *For $\alpha > 0$ and real x , with F extended as a distribution function,*

$$(I_{0+}^{\alpha}F)(x) = \frac{1}{\Gamma(\alpha+1)} \mathbb{E}(x-X)_{+}^{\alpha}. \quad (7.3)$$

For $0 < \alpha < 1$ and $0 < x < 1$,

$$(D_{0+}^{\alpha}F)(x) = ({}^C D_{0+}^{\alpha}F)(x) = \frac{1}{\Gamma(1-\alpha)} \mathbb{E}(x-X)_{+}^{-\alpha}. \quad (7.4)$$

Proved in this report.

Proof. Insert $F(t) = \mathbb{E}1_{\{X \leq t\}}$ into (7.1) and interchange expectation and integration. The inner integral from X to x equals $(x-X)^{\alpha}/\alpha$. For $0 < \alpha < 1$, differentiate the formula with α replaced by $1-\alpha$. The local singularity is integrable because the density of X is continuous. Equality with the Caputo derivative uses $F(0) = 0$. $\square$

The random fixed point (2.6) gives a fractional refinement law which interpolates the ordinary repeated-integration hierarchy.

Theorem 7.2 (Fractional Fabius refinement). *Let $J_{\alpha}(x) = (I_{0+}^{\alpha}F)(x)$, with $J_{\alpha}(x) = 0$ for $x < 0$. Then for every $\alpha > 0$,*

$$J_{\alpha}(x) = 2^{-\alpha} (J_{\alpha+1}(2x) - J_{\alpha+1}(2x-1)). \quad (7.5)$$

On $0 < x < 1/2$, the corresponding derivative scaling is

$$D_{0+}^{\alpha+1}F(x) = 2^{\alpha+1} (D_{0+}^{\alpha}F)(2x), \quad (7.6)$$

whenever interpreted classically, and otherwise in the distributional sense. *Proved in this report.*

Proof. From $X \stackrel{d}{=} (U + X')/2$,

$$\mathbb{E}(x-X)_{+}^{\alpha} = 2^{-\alpha} \mathbb{E}(2x-U-X')_{+}^{\alpha}.$$

Averaging first over $U \in [0, 1]$ gives the difference of two $(\alpha+1)$ -st truncated powers divided by $\alpha+1$, proving (7.5). The derivative law follows from $F'(x) = 2F(2x)$, the scaling rule for left fractional derivatives, and endpoint flatness. $\square$

7.3 Weighted fractional antiderivatives: the incomplete-beta master theorem

For $a, b > 0$, write

$$B_z(a, b) = \int_0^z t^{a-1} (1-t)^{b-1} dt. \quad (7.7)$$

Theorem 7.3 (Fractional monomial–Fabius/up integrals). *Let $\alpha > 0$, $\Re p > -1$, and $0 < x \leq 1$. Then*

$$I_{0+}^\alpha [t^p \text{up}(t)](x) = \frac{x^{p+\alpha}}{\Gamma(\alpha)} \mathbb{E}[B_{\min(X/x, 1)}(p+1, \alpha)], \quad (7.8)$$

$$I_{0+}^\alpha [t^p F(t)](x) = \frac{x^{p+\alpha}}{\Gamma(\alpha)} \mathbb{E}[\mathbf{1}_{\{X < x\}} B_{1-X/x}(\alpha, p+1)]. \quad (7.9)$$

At $x = 1$,

$$I_{0+}^\alpha [t^p \text{up}(t)](1) = \frac{1}{\Gamma(\alpha)} \sum_{j=0}^{\infty} \frac{(1-\alpha)_j}{j! (p+j+1)} m(p+j+1), \quad (7.10)$$

$$I_{0+}^\alpha [t^p F(t)](1) = \frac{\Gamma(p+1)}{\Gamma(p+\alpha+1)} - I_{0+}^\alpha [t^p \text{up}(t)](1). \quad (7.11)$$

The series converges absolutely for $\alpha > 0$. If $\alpha = n \in \mathbb{N}$, it terminates:

$$I_{0+}^n [t^p \text{up}(t)](1) = \frac{1}{(n-1)!} \sum_{j=0}^{n-1} (-1)^j \binom{n-1}{j} \frac{m(p+j+1)}{p+j+1}. \quad (7.12)$$

Proved in this report.

Proof. For the up-function, insert $\text{up}(t) = \mathbb{E}\mathbf{1}_{\{t < X\}}$ into the Riemann–Liouville integral. The inner range is $0 < t < \min(X, x)$; scaling $t = xr$ gives (7.8). For F , the inner range is $X < t < x$, producing the complementary incomplete beta integral in (7.9).

At $x = 1$, expand $(1-t)^{\alpha-1} = \sum_{j \geq 0} (1-\alpha)_j t^j / j!$ in $B_X(p+1, \alpha)$, then take expectations. The coefficients are $\mathcal{O}(j^{-\alpha})$, the extra denominator is j^{-1} , and $|m(p+j+1)| \leq 1$ in the initial domain, giving absolute convergence. For integer α , the Pochhammer symbol truncates. $\square$

Remark 7.4 (Why this answers several integral questions at once). Setting $\alpha = 1$ recovers the ordinary antiderivatives (3.3)–(3.4). Integer $\alpha > 1$ gives every repeated antiderivative as a finite combination of generalized moments. Noninteger α gives a convergent moment series. Negative and fractional p are allowed down to the sharp barrier $\Re p = -1$ for the up-integral; the Fabius integral itself continues beyond that barrier because of endpoint flatness.

7.4 Fractional primitives of the up-function

Define, for $\alpha > 0$,

$$H_\alpha(x) = (I_{-1+}^\alpha \text{up})(x) = \frac{1}{\Gamma(\alpha)} \mathbb{E}(x - Y)_+^{\alpha-1}. \quad (7.13)$$

Theorem 7.5 (Fractional up-refinement). *For all $\alpha > 0$ and real x ,*

$$H_\alpha(x) = 2^{-\alpha} (H_{\alpha+1}(2x+1) - H_{\alpha+1}(2x-1)). \quad (7.14)$$

For $x > 1$,

$$H_\alpha(x) = \frac{x^{\alpha-1}}{\Gamma(\alpha)} \sum_{r=0}^{\infty} \binom{\alpha-1}{2r} \mu_{2r} x^{-2r}. \quad (7.15)$$

The exterior series converges absolutely; for integer α it reduces to (6.10). *Proved in this report.*

Proof. Use $Y \stackrel{d}{=} (V + Y')/2$ and average first over uniform $V \in [-1, 1]$. The integral of $(2x - Y' - V)_+^{\alpha-1}$ over V produces the difference of two α -th powers and the factor $2^{-\alpha}$. For $x > 1$, expand $(1 - Y/x)^{\alpha-1}$; symmetry removes all odd moments. $\square$

For general derivative order $n - 1 < \alpha < n$, the one-sided Riemann–Liouville derivative is exactly

$$D_{-1+}^\alpha \text{up}(x) = \frac{d^n}{dx^n} H_{n-\alpha}(x). \quad (7.16)$$

This formula is computationally useful because [theorem 7.5](#) reduces every H_β to scaled copies one order higher.

7.5 Fractional calculus of the inverse Fabius function

Theorem 7.6 (Fractional inverse layer cake). *For $\alpha > 0$ and $0 \leq u \leq 1$,*

$$(I_{0+}^\alpha Q)(u) = \frac{1}{\Gamma(\alpha + 1)} \int_0^{Q(u)} (u - F(x))^\alpha dx. \quad (7.17)$$

At the endpoint,

$$\int_0^1 \text{up}(x)^\alpha dx = \Gamma(\alpha + 1) (I_{0+}^\alpha Q)(1). \quad (7.18)$$

For $0 < \alpha < 1$,

$$(D_{0+}^\alpha Q)(u) = ({}^C D_{0+}^\alpha Q)(u) = \frac{1}{\Gamma(1 - \alpha)} \int_0^{Q(u)} (u - F(x))^{-\alpha} dx. \quad (7.19)$$

Proved in this report.

Proof. Apply [theorem 5.2](#) with $\phi(v) = (u - v)_+^{\alpha-1}/\Gamma(\alpha)$ and $\Psi(x) = x$. At $u = 1$, use $1 - F(x) = \text{up}(x)$. For the derivative, differentiate the formula for $I_{0+}^{1-\alpha} Q$. The moving-boundary term vanishes because its positive power is zero at $x = Q(u)$, and the remaining singularity is integrable for $\alpha < 1$. Since $Q(0) = 0$, the Riemann–Liouville and Caputo derivatives agree. $\square$

7.6 A full-line Riesz refinement law

Let D_R^α be the full-line Riesz derivative with Fourier multiplier $|2\pi\xi|^\alpha$. Applying it to [\(2.7\)](#), and using the exact scaling law, gives

$$\frac{d}{dx} (D_R^\alpha \text{up})(x) = 2^{\alpha+1} [(D_R^\alpha \text{up})(2x + 1) - (D_R^\alpha \text{up})(2x - 1)]. \quad (7.20)$$

The identity holds in tempered distributions for every $\alpha \geq 0$, and classically wherever the Riesz derivative is represented by an ordinary function. It is the fractional-order analogue of the differential refinement equation with no ambiguity from a moving one-sided terminal.

8 Cauchy, Stieltjes, and Hilbert transforms

8.1 The Cauchy transform and its representations

Define

$$S(z) = \int_{-1}^1 \frac{\text{up}(t)}{z - t} dt, \quad z \in \mathbb{C} \setminus [-1, 1]. \quad (8.1)$$

Since up is even and has moments μ_{2r} ,

$$S(z) = \sum_{r=0}^{\infty} \frac{\mu_{2r}}{z^{2r+1}}, \quad |z| > 1. \quad (8.2)$$

For $\Re z > 1$, the exponential resolvent gives

$$S(z) = \int_0^{\infty} e^{-zs} \Phi(s) \, ds. \quad (8.3)$$

The random fixed point $Y = (V + Y')/2$ also gives the nonlinear logarithmic identity

$$S(z) = \int_{-1}^1 \text{up}(y) \log\left(\frac{2z - y + 1}{2z - y - 1}\right) \, dy, \quad (8.4)$$

with a branch chosen consistently off $[-1, 1]$.

8.2 An exact functional-differential equation

Theorem 8.1 (Cauchy renormalization equation). *The Cauchy transform of the up-law satisfies*

$$S'(z) = 2(S(2z + 1) - S(2z - 1)), \quad z \notin [-1, 1]. \quad (8.5)$$

Proved in this report.

Proof. Differentiate under the integral and integrate by parts, using the flat endpoint values $\text{up}(\pm 1) = 0$:

$$S'(z) = - \int_{-1}^1 \frac{\text{up}(t)}{(z - t)^2} \, dt = \int_{-1}^1 \frac{\text{up}'(t)}{z - t} \, dt.$$

Insert (2.7). In the first term use $y = 2t + 1$; in the second use $y = 2t - 1$. The two transformed integrals are exactly $S(2z + 1)$ and $S(2z - 1)$. $\square$

Expanding (8.5) at infinity reproduces the rational moment recurrence. Conversely, the equation plus $S(z) \sim 1/z$ packages the whole moment sequence into one analytic renormalization law.

For the Fabius density, define

$$T(z) = \int_0^1 \frac{F'(x)}{z - x} \, dx = 2S(2z - 1). \quad (8.6)$$

Then

$$T'(z) = 4(T(2z) - T(2z - 1)). \quad (8.7)$$

The positive-half up resolvent has a particularly simple expectation form:

$$\int_0^1 \frac{\text{up}(x)}{z - x} \, dx = \mathbb{E} \log \frac{z}{z - X}. \quad (8.8)$$

8.3 Boundary values and the Hilbert transform

For $-1 < x < 1$, the Sokhotski–Plemelj boundary values are

$$S(x + i0) = \text{p. v.} \int_{-1}^1 \frac{\text{up}(t)}{x - t} \, dt - i\pi \text{up}(x). \quad (8.9)$$

Thus the real part is π times the Hilbert transform under the convention

$$(\mathcal{H} \text{up})(x) = \frac{1}{\pi} \text{p. v.} \int_{-1}^1 \frac{\text{up}(t)}{x - t} \, dt. \quad (8.10)$$

Taking nontangential limits in (8.5) yields coupled refinement equations for $\mathcal{H} \text{up}$ and up , with the expected changes when $2x \pm 1$ enters or leaves the support. This gives a systematic route to boundary-value computation without direct principal-value quadrature.

9 The Jacobi continued fraction of the up-law

9.1 From rational moments to a J-fraction

Let P_n be the monic orthogonal polynomial of degree n for the probability measure $\text{up}(x) dx$. Existing corpus work proves positivity of the Hankel determinants and develops these polynomials for Gaussian quadrature [5]. Evenness forces the diagonal recurrence parameters to vanish, so

$$P_{n+1}(x) = xP_n(x) - a_n P_{n-1}(x), \quad P_0 = 1, \quad P_1 = x, \quad (9.1)$$

where

$$a_n = \frac{h_n}{h_{n-1}} = \frac{\Delta_{n+1}\Delta_{n-1}}{\Delta_n^2} > 0. \quad (9.2)$$

Here $h_n = \int P_n^2 \text{up}$ and $\Delta_n = \det(\mu_{i+j})_{0 \leq i,j < n}$. Rational moments imply $a_n \in \mathbb{Q}_{>0}$.

Theorem 9.1 (Symmetric Jacobi fraction). *The Cauchy transform S has the locally uniform continued fraction*

$$S(z) = \frac{1}{z - \frac{a_1}{z - \frac{a_2}{z - \frac{a_3}{z - \ddots}}}}, \quad z \in \mathbb{C} \setminus [-1, 1]. \quad (9.3)$$

All coefficients are positive rationals. The first exact values are displayed in [table 1](#). Moreover,

$$\lim_{n \rightarrow \infty} a_n = \frac{1}{4}. \quad (9.4)$$

Proved in this report.

Proof. The Stieltjes–Jacobi correspondence for a compactly supported positive measure gives (9.3); symmetry gives zero diagonal parameters, and (9.2) gives positivity and rationality. The up-density is strictly positive almost everywhere on $(-1, 1)$. Rakhmanov’s theorem for orthogonal polynomials on $[-1, 1]$ therefore gives orthonormal off-diagonal coefficients tending to $1/2$; squaring converts them to the monic coefficients $a_n \rightarrow 1/4$ [14]. $\square$

Table 1: First Jacobi coefficients of the monic up-polynomials, computed by exact rational Gram–Schmidt.

n	a_n (exact)	decimal
1	$\frac{1}{9}$	0.111111111111
2	$\frac{32}{225}$	0.142222222222
3	$\frac{619}{3675}$	0.168435374150
4	$\frac{835232}{4640643}$	0.179981955087
5	$\frac{1789043898625}{9272274723657}$	0.192945523288

The first polynomials are

$$P_0(x) = 1, \quad P_1(x) = x, \quad (9.5)$$

$$P_2(x) = x^2 - \frac{1}{9}, \quad P_3(x) = x^3 - \frac{19}{75}x, \quad (9.6)$$

$$P_4(x) = x^4 - \frac{62}{147}x^2 + \frac{619}{33075}. \quad (9.7)$$

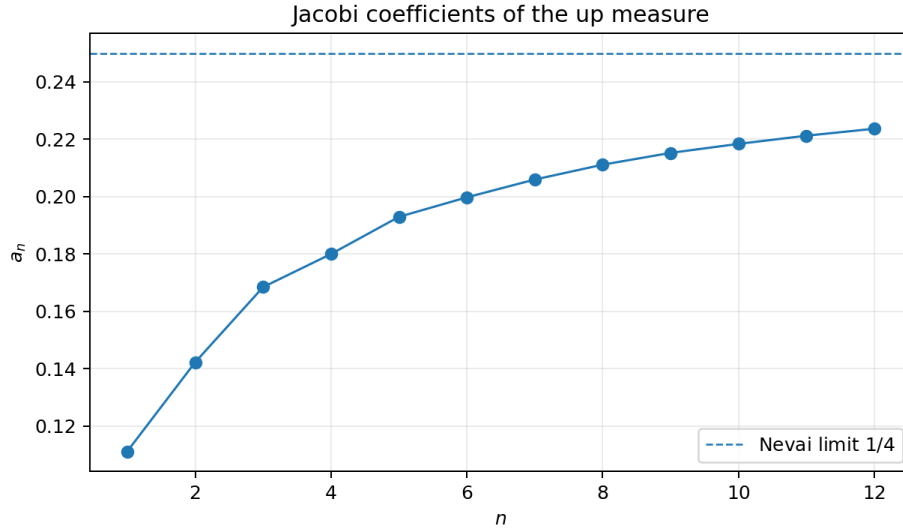


Figure 1: The first twelve exact Jacobi coefficients (shown in decimal form) and the rigorous limiting value $1/4$. Monotonicity is suggested but not proved.

The convergents of (9.3) are diagonal Padé approximants to the moment expansion (8.2). Hence the functional equation (8.5) can be used to test or accelerate rational approximants across dyadic affine scales.

Conjecture 9.2 (Monotone Nevai approach). *The monic Jacobi coefficients of the up-law satisfy*

$$0 < a_1 < a_2 < a_3 < \cdots < \frac{1}{4}. \quad (9.8)$$

Conjectural.

The first twelve coefficients support the conjecture, but endpoint flatness can produce subtle high-order recurrence effects. A proof may require total positivity of suitable Hankel-ratio arrays, a comparison theorem for even weights, or a direct use of (8.5) in Jacobi coordinates.

10 Nonlinear powers and order statistics

10.1 Exact order-statistic identities

Let $X_1, X_2, \dots$ be independent copies of X , and let $Q = F^{-1}$.

Theorem 10.1 (Power integrals as minima). *For every integer $p \geq 1$,*

$$\int_0^1 \text{up}(x)^p dx = \int_0^1 F(x)^p dx = \mathbb{E} \min(X_1, \dots, X_p). \quad (10.1)$$

Equivalently,

$$\int_0^1 \text{up}(x)^p dx = p \int_0^1 Q(v)(1-v)^{p-1} dv. \quad (10.2)$$

Proved in this report.

Proof. The survival function of the minimum is $\text{up}(x)^p$, so its expectation is the first integral. Reflection gives $\int F^p = \int \text{up}^p$. If $B_p = \min(U_1, \dots, U_p)$ for uniform variables, then B_p has density $p(1-v)^{p-1}$, and $\min(X_1, \dots, X_p) \stackrel{d}{=} Q(B_p)$. $\square$

Theorem 10.2 (Mixed powers as a random gap). *Let $m, n \geq 1$, let $A = \max(X_1, \dots, X_m)$, and let $B = \min(X_{m+1}, \dots, X_{m+n})$. Then*

$$\int_0^1 F(x)^m \text{up}(x)^n dx = \mathbb{E}(B - A)_+. \quad (10.3)$$

Proved in this report.

Proof. For fixed x , $F(x)^m \text{up}(x)^n = \mathbb{P}(A \leq x < B)$. Integrate the indicator over $x \in [0, 1]$ and use Fubini. $\square$

Fractional exponents are linked to the inverse without introducing a noninteger number of samples: (7.18) gives

$$\int_0^1 \text{up}(x)^\alpha dx = \Gamma(\alpha + 1)(I_{0+}^\alpha Q)(1), \quad \alpha > 0. \quad (10.4)$$

10.2 Large-power asymptotics from inverse slow variation

The inverse-frontier corpus proves the endpoint elasticity [6]

$$\frac{uQ'(u)}{Q(u)} \rightarrow 0 \quad (u \downarrow 0), \quad (10.5)$$

with a sharper lower-Lambert and periodic expansion. It follows that Q is slowly varying at zero:

$$\frac{Q(cu)}{Q(u)} \rightarrow 1 \quad (u \downarrow 0), \quad c > 0. \quad (10.6)$$

Theorem 10.3 (Power-integral scale). *As $p \rightarrow \infty$ through positive integers,*

$$\int_0^1 \text{up}(x)^p dx \sim Q(1/p). \quad (10.7)$$

Proved in this report.

Proof. From (10.2), substitute $v = t/p$:

$$\frac{\int_0^1 \text{up}(x)^p dx}{Q(1/p)} = \int_0^p \frac{Q(t/p)}{Q(1/p)} \left(1 - \frac{t}{p}\right)^{p-1} dt.$$

For fixed $t > 0$, slow variation gives convergence of the ratio to one, and the beta kernel converges to e^{-t} . Potter bounds for slowly varying functions provide an integrable majorant after splitting at $t = 1$; the far tail is exponentially negligible. Dominated convergence gives $\int_0^\infty e^{-t} dt = 1$. $\square$

Define the small elasticity

$$h_p = \frac{p^{-1}Q'(p^{-1})}{Q(p^{-1})}. \quad (10.8)$$

The endpoint theory gives $h_p \rightarrow 0$, at a logarithmically slow rate.

Proposition 10.4 (Conditional second-order law). *Let B_p have the Beta(1, p) law. Assume the local self-neglecting property*

$$\frac{tuQ'(tu)/Q(tu)}{uQ'(u)/Q(u)} \rightarrow 1 \quad (10.9)$$

uniformly for t in compact subsets of $(0, \infty)$, and assume that the family

$$Z_p = \frac{Q(B_p)/Q(1/p) - 1}{h_p} \quad (10.10)$$

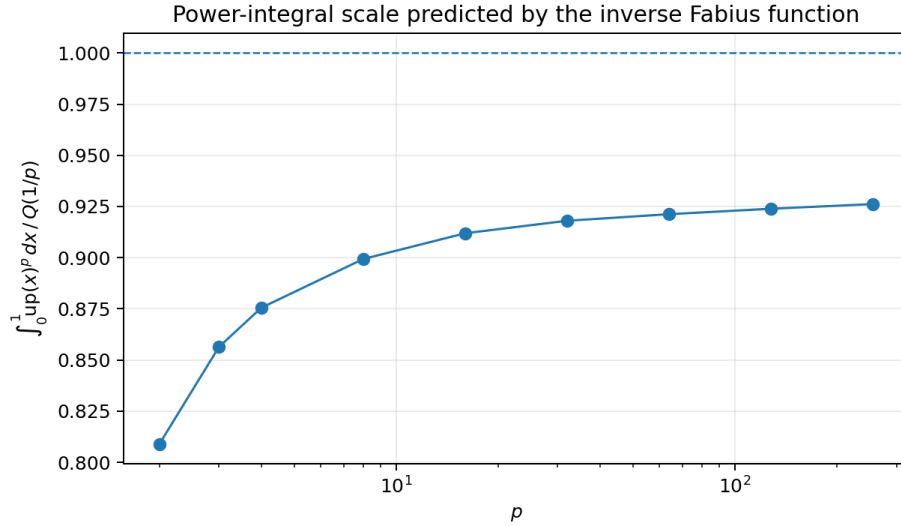


Figure 2: Numerical approach of $\int_0^1 \text{up}(x)^p dx / Q(1/p)$ toward the rigorous limit one. The slow approach reflects the lower-Lambert endpoint scale.

is uniformly integrable. Then

$$\int_0^1 \text{up}(x)^p dx = Q(1/p)(1 - \gamma h_p + o(h_p)), \quad (10.11)$$

where γ is Euler's constant. More precisely,

$$Z_p \implies \log E, \quad E \sim \text{Exp}(1). \quad (10.12)$$

A second-order Potter bound dominating Z_p is one sufficient replacement for the stated uniform-integrability assumption. **Conditional statement.**

Proof under the stated hypotheses. Since $pB_p \Rightarrow E$, integrate the logarithmic derivative of Q between $1/p$ and t/p . The self-neglecting condition gives

$$\log \frac{Q(t/p)}{Q(1/p)} = h_p \log t + o(h_p)$$

locally uniformly. Exponentiation yields the distributional limit. The stated uniform integrability permits passage to expectations, and $\mathbb{E} \log E = -\gamma$ gives (10.11). $\square$

Conjecture 10.5 (Second-order endpoint regularity). *The self-neglecting condition (10.9) and the required uniform integrability follow from the repository's all-orders inverse lower-Lambert expansion, uniformly on every compact multiplicative t -range. Consequently (10.11) holds without an additional hypothesis.* **Conjectural.**

11 Correlation and transform families

Although autocorrelation identities are standard Fourier consequences, they provide useful definite integrals involving two copies of the up-function. Define

$$C(a) = \int_{\mathbb{R}} \text{up}(x) \text{up}(x+a) dx = (\text{up} * \text{up})(a). \quad (11.1)$$

With the Fourier convention $\hat{f}(\xi) = \int f(x) e^{-2\pi i x \xi} dx$,

$$\hat{C}(\xi) = |\widehat{\text{up}}(\xi)|^2 = \prod_{j=0}^{\infty} \text{sinc}^2\left(\frac{\pi \xi}{2^j}\right). \quad (11.2)$$

Thus

$$C(a) = \int_{\mathbb{R}} \prod_{j=0}^{\infty} \text{sinc}^2\left(\frac{\pi \xi}{2^j}\right) e^{2\pi i a \xi} d\xi. \quad (11.3)$$

More generally,

$$\int_{\mathbb{R}} \text{up}^{(m)}(x) \text{up}^{(n)}(x+a) dx = \int_{\mathbb{R}} (-2\pi i \xi)^m (2\pi i \xi)^n |\widehat{\text{up}}(\xi)|^2 e^{2\pi i a \xi} d\xi. \quad (11.4)$$

At $a = 0$,

$$\|\text{up}^{(m)}\|_2^2 = (2\pi)^{2m} \int_{\mathbb{R}} \xi^{2m} \prod_{j=0}^{\infty} \text{sinc}^2\left(\frac{\pi \xi}{2^j}\right) d\xi. \quad (11.5)$$

These formulae turn Sobolev energies and shifted derivative overlaps into integral transforms of the squared infinite sinc product. The corpus's Fourier decay theory supplies convergence at every derivative order.

12 Numerical verification and reproducibility

12.1 Exact and floating-point layers

The companion program `numerical_experiments.py` separates proof from experiment.

1. Bernoulli cumulants, moments, Gram polynomials, and Jacobi coefficients are computed with Python's exact `Fraction` arithmetic.
2. A uniform grid on $[-1, 1]$ is iterated under the positive fixed-point operator

$$(Tg)(x) = \int_{2x-1}^{2x+1} g(t) dt.$$

The resulting density is used only to cross-check analytic identities.

3. Adaptive or grid quadrature checks the Cauchy equation, fractional refinement, inverse fractional layer cake, order-statistic formulae, and generalized-moment Newton series.

With 65,537 grid points and 25 refinement iterations, the main diagnostics were

Check	observed residual / error
Mass of reconstructed up	2.2×10^{-16} from one
Variance versus exact $1/9$	5.2×10^{-11}
Maximum symmetry defect	1.5×10^{-14}
Cauchy equation at three complex points	relative 2.9×10^{-11} to 5.6×10^{-11}
Fractional up-refinement	1.3×10^{-12} to 7.7×10^{-7}
Inverse fractional identity	9.4×10^{-11} to 1.3×10^{-8}
Newton series for $m(-1/2), m(1/2), m(3/2)$	3.3×10^{-8} to 8.0×10^{-12}
Logarithmic identity $\int F(x)/x dx = -m'(0)$	4.4×10^{-9}

The full exact fractions through a_{12} , every test point, precision setting, and residual are recorded in `experiments_summary.txt`. The script is deterministic and uses no Monte Carlo simulation.

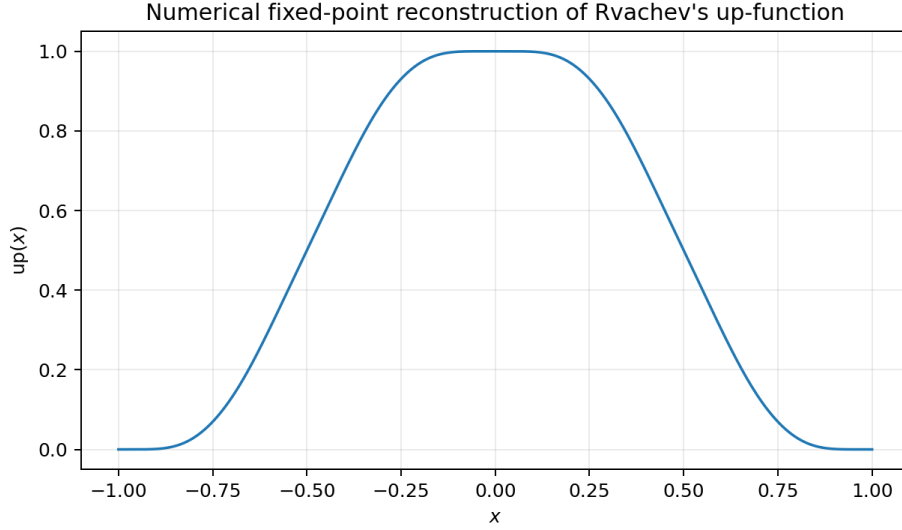


Figure 3: Positive fixed-point reconstruction used for numerical cross-checks. The exact theory, not this discretization, is the source of all theorems.

12.2 Reproduction command

From the extracted archive directory:

```
python numerical_experiments.py --output-dir reproduced
latexmk -pdf -interaction=nonstopmode -halt-on-error \
  Fabius_Integral_Transform_Fractional_Frontiers.tex
```

The generated table `jacobi_coefficients.tex` and all three figures are included in the PDF build.

13 Conjectures and future research directions

13.1 Cauchy renormalization as a spectral equation

Equation (8.5) deserves treatment as a boundary-value renormalization problem in its own right.

Frontier problem 13.1 (Uniqueness from analytic data). Characterize the weakest growth and Herglotz conditions under which

$$S'(z) = 2[S(2z + 1) - S(2z - 1)], \quad S(z) \sim z^{-1},$$

uniquely determines the up-law. A positive answer would provide a transform- space definition of up independent of the infinite convolution.

Frontier problem 13.2 (Dyadic Padé acceleration). Combine the J-fraction convergents with the affine values $S(2z \pm 1)$ in (8.5) to build certified rational approximants on $\mathbb{C} \setminus [-1, 1]$. Determine whether the dyadic equation improves the usual Markov convergence bounds for Stieltjes continued fractions.

13.2 Jacobi arithmetic and endpoint flatness

The exact rational coefficients a_n form a new arithmetic sequence attached to the up-law. Beyond theorem 9.2, natural questions are:

1. determine sharp denominator-clearing products in terms of Mersenne factors $2^k - 1$ or $4^k - 1$;
2. find the first correction to $a_n = 1/4 + o(1)$, and decide whether endpoint flatness produces a logarithmic, stretched-exponential, or oscillatory rate;

3. translate the Cauchy functional equation into a closed hierarchy for Jacobi parameters or Hankel ratios;
4. construct a Lean proof of the first exact coefficients from the already formalized rational moment recurrence.

13.3 Fractional endpoint asymptotics

The repository's endpoint expansion controls $F(x)$ and $Q(u)$ through a lower-Lambert variable and periodic Gamma–zeta terms. The expectation kernels in (7.3), (7.9), and (7.17) suggest a larger transseries.

Conjecture 13.3 (Fractional lower-Lambert transseries). *For fixed noninteger $\alpha > 0$, the endpoint asymptotic expansions of $I_{0+}^\alpha F$, $D_{0+}^\alpha F$, and $I_{0+}^\alpha Q$ are organized by the same exact lower-Lambert phase as F and Q , with coefficients obtained by beta/gamma shifts of the existing periodic saddle jets. **Conjectural.***

A practical route is to insert the known endpoint density expansion into the truncated-power expectations, split at the saddle scale, and carry the beta kernel through the existing uniform remainder estimates.

13.4 Nonlinear power integrals

The first-order equivalence (10.7) is rigorous. The next frontier is to prove the self-neglecting property required in theorem 10.4 and then propagate the complete inverse transseries through the beta kernel. This should produce

$$\int_0^1 \text{up}(x)^p dx = Q(1/p) (1 - \gamma h_p + c_2 h_p^2 + \text{extperiodiccorrections} + \dots), \quad (13.1)$$

where c_2 should involve moments of $\log E$, hence $\pi^2/6$ and γ^2 , while the periodic terms inherit the Gamma–zeta phase of the inverse Fabius expansion.

13.5 Fractional formalization roadmap

Formalization obligation 13.4 (Kernel layer). Formalize Tonelli identities for survival functions and quantiles, including theorems 3.1 and 5.2. These are measure-theoretic but structurally short.

Formalization obligation 13.5 (Special-function layer). Introduce incomplete beta integrals and prove theorem 7.3 first for positive real parameters. Complex continuation can remain outside the initial formalization.

Formalization obligation 13.6 (Transform layer). Build the Cauchy transform off a compact interval, justify differentiation and integration by parts, and formalize (8.5). The proof uses only the already established up refinement equation and compact support.

Formalization obligation 13.7 (Orthogonal-polynomial layer). Reuse the corpus's Hankel-positivity and Gaussian-quadrature infrastructure to construct $a_n = h_n/h_{n-1}$, the finite J-fraction convergents, and exact certificates for the first coefficients. Rakhmanov's asymptotic theorem may be imported as a separately documented analytic dependency.

14 Conclusion

The Fabius–Rvachev system admits a broader integral calculus than its original differential and Fourier definitions suggest. Three probability identities supply the organizing principle:

$$\text{up}(x) = \mathbb{P}(X > x), \quad F(x) = \mathbb{P}(X \leq x), \quad Q(u) = \inf\{x : F(x) \geq u\}. \quad (14.1)$$

From them follow:

1. arbitrary-kernel and monomial antiderivatives as truncated moments;
2. an entire generalized-moment function reconstructed from rational moments by a Newton series;
3. a Mellin triad distinguishing the single pole of up from the entire transforms of F and F' ;
4. exact layer-cake transforms and fractional derivatives of the inverse;
5. incomplete-beta formulae interpolating ordinary, repeated, negative-power, and fractional antiderivatives;
6. a Cauchy functional-differential equation and the explicit Jacobi fraction requested by an existing frontier report;
7. order-statistic interpretations and a sharp inverse-Fabius scale for nonlinear power integrals.

The most promising next step is to combine the new kernel formulae with the repository's all-orders lower-Lambert endpoint theory. That synthesis should turn several conditional statements here—especially the second-order power integral and fractional endpoint transseries—into explicit theorems with periodic Gamma–zeta coefficients.

A Exact formula sheet

Object	Exact formula
Fabius moment	$m(s) = \mathbb{E}X^s = \sum_{n \geq 0} (-1)^n \binom{s}{n} m_n$
Mellin of positive up-half	$\int_0^1 x^{s-1} \text{up}(x) dx = m(s)/s$
Mellin of F	$\int_0^1 x^{s-1} F(x) dx = (1 - m(s))/s$
Monomial up integral	$\int_0^1 x^p \text{up}(x) dx = m(p+1)/(p+1), \Re p > -1$
Monomial Fabius integral	$\int_0^1 x^p F(x) dx = (1 - m(p+1))/(p+1), \text{entire in } p$
Laplace/exponential up-half	$\int_0^1 \text{up}(x) e^{zx} dx = (A(z) - 1)/z$
Laplace/exponential Fabius	$\int_0^1 F(x) e^{zx} dx = (e^z - A(z))/z$
Quantile moments	$\int_0^1 Q(u)^p du = m(p)$
Inverse primitive	$\int_0^u Q(v) dv = uQ(u) - \int_0^{Q(u)} F(x) dx$
Fractional F primitive	$I_{0+}^\alpha F(x) = \mathbb{E}(x - X)_+^\alpha / \Gamma(\alpha + 1)$
Fractional inverse primitive	$I_{0+}^\alpha Q(u) = \Gamma(\alpha + 1)^{-1} \int_0^{Q(u)} (u - F(x))^\alpha dx$
Power/fractional bridge	$\int_0^1 \text{up}(x)^\alpha dx = \Gamma(\alpha + 1) I_{0+}^\alpha Q(1)$
Cauchy equation	$S'(z) = 2[S(2z + 1) - S(2z - 1)]$
Jacobi fraction	$S(z) = 1/(z - a_1/(z - a_2/(z - \dots))), a_n \in \mathbb{Q}_{>0}, a_n \rightarrow 1/4$
Power asymptotic	$\int_0^1 \text{up}(x)^p dx \sim Q(1/p)$

B Corpus families inspected

The recursive audit was organized by mathematical lineage. The principal families were:

1. the living primary exposition *Fabius Function and Rvachev Up*;
2. the mathematical glossary, non-elementarity article, and Lean walkthrough;
3. the consolidated non-formalized frontier volume and provenance/gap register;
4. repeated-integration and up-from-integration manuscripts;
5. dyadic formula, q -connection, asymptotic bridge, and small-argument manuscripts;

6. inverse-function, inverse-saddle, and lower-Lambert reports;
7. the Thue–Morse formula atlas and its archived drafts;
8. active arithmetic-ray, self-sampling, half-integer spectral, exponent- sequence/Newton, spectral-Pascal, finite-block, sinc-Richardson, and Fourier-decay reports;
9. imported TeX sources for the classical Fabius, Rvachev, arithmetic, and compact-support literature;
10. frozen archive variants retained by the repository’s provenance map.

The exact 67-path list at the pinned commit is recorded in the inventory appendix of the exponent-sequence/Newton report [7]. Current descendants and newly active draft folders were rechecked separately so that the audit did not stop at the pinned snapshot.

C Reproducibility manifest

The companion ZIP archive contains:

Path	Purpose
Fabius_Integral_Transform_Fractal_Frontiers.tex	Complete LaTeX manuscript
Fabius_Integral_Transform_Fractal_Frontiers.pdf	Compiled and visually inspected report
numerical_experiments.py	Commented exact/numerical verification program
experiments_summary.txt	Full deterministic diagnostics and exact coefficients
jacobi_coefficients.tex	Generated exact table used in the manuscript
up_reconstruction.png	Fixed-point density diagnostic
jacobi_coefficients.png	Exact-coefficient trend figure
power_quantile_ratio.png	Power-integral asymptotic diagnostic
requirements.txt	Python package versions used
README.md	Build and reproduction instructions
SHA256SUMS	Checksums for packaged artifacts

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